7. Demonstration of clustering rule process on dataset student.arff using hierarchical clustering.

Description:

Hierarchical clustering involves creating **clusters** that have a predetermined ordering from top to bottom. For example, all files and folders on the hard disk are organized in a **hierarchy**. There are two types of **hierarchical clustering**, Divisive and Agglomerative.

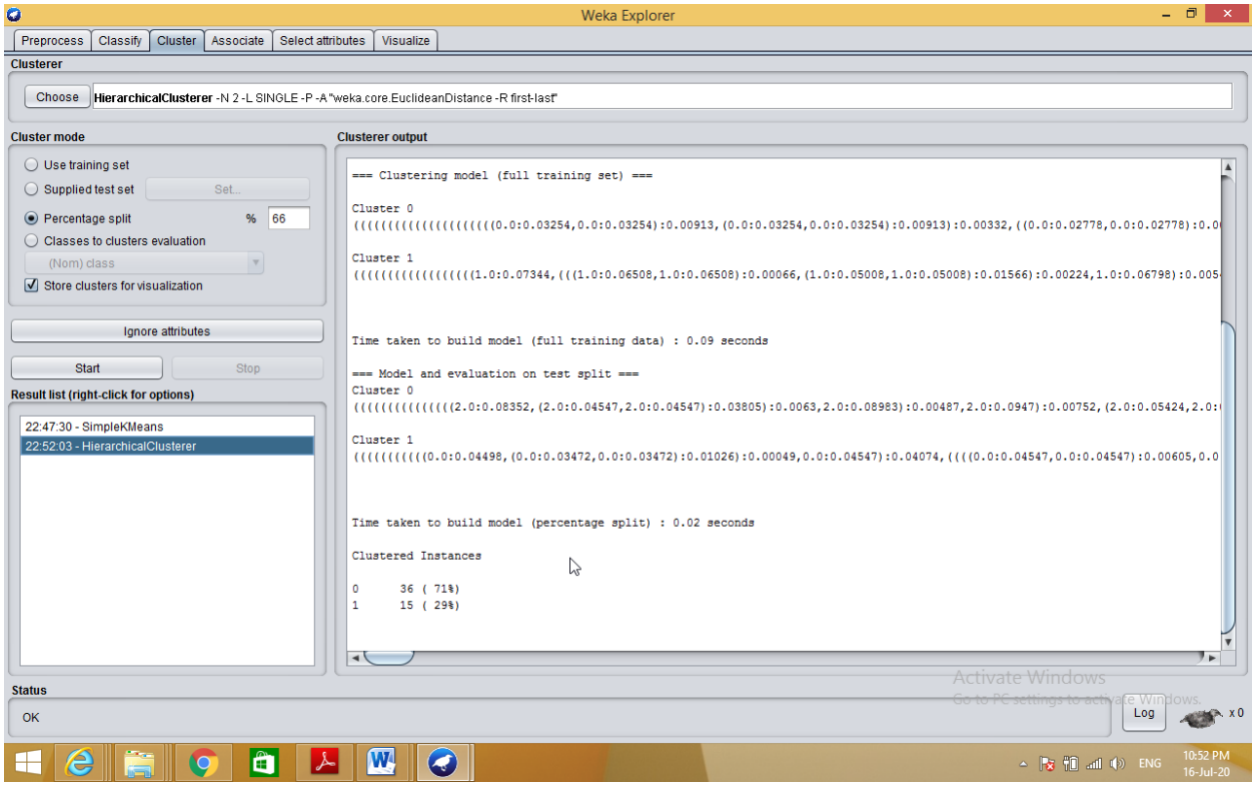
Procedure:

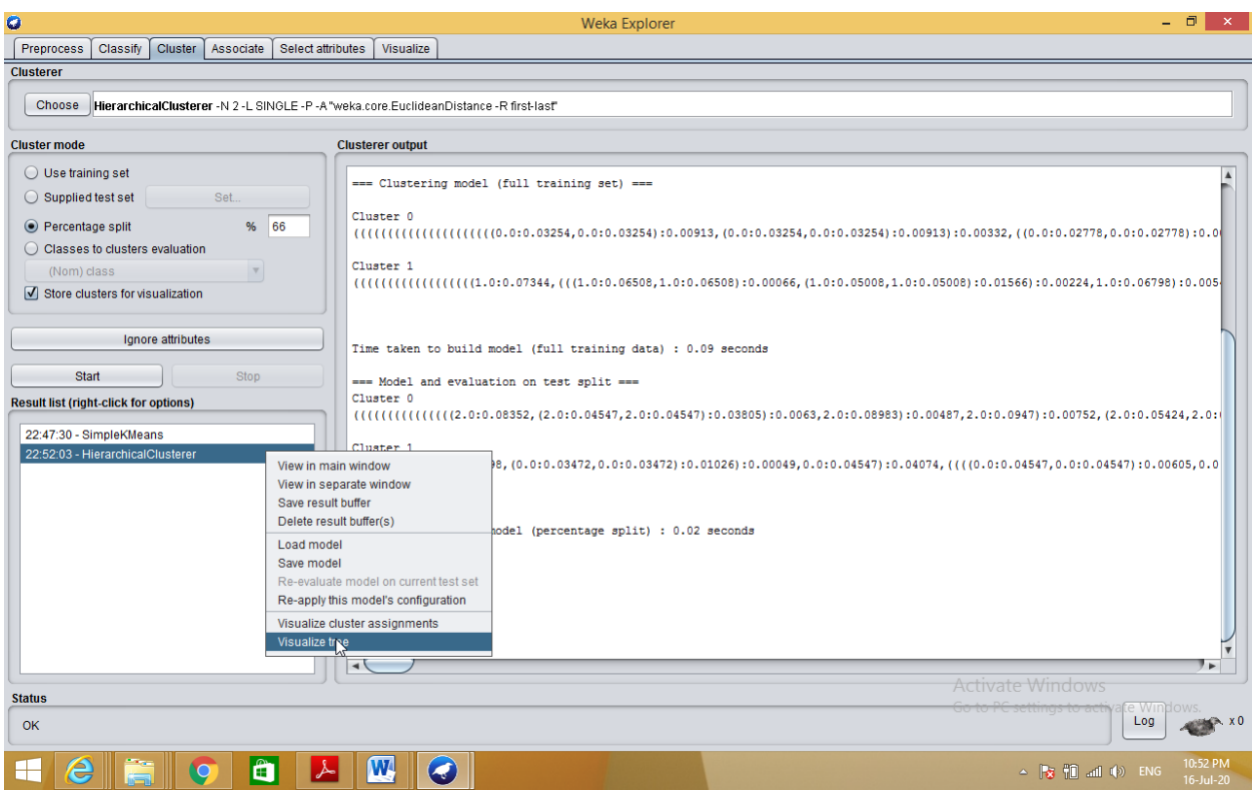
Step 1: Load appropriate dataset into WEKA

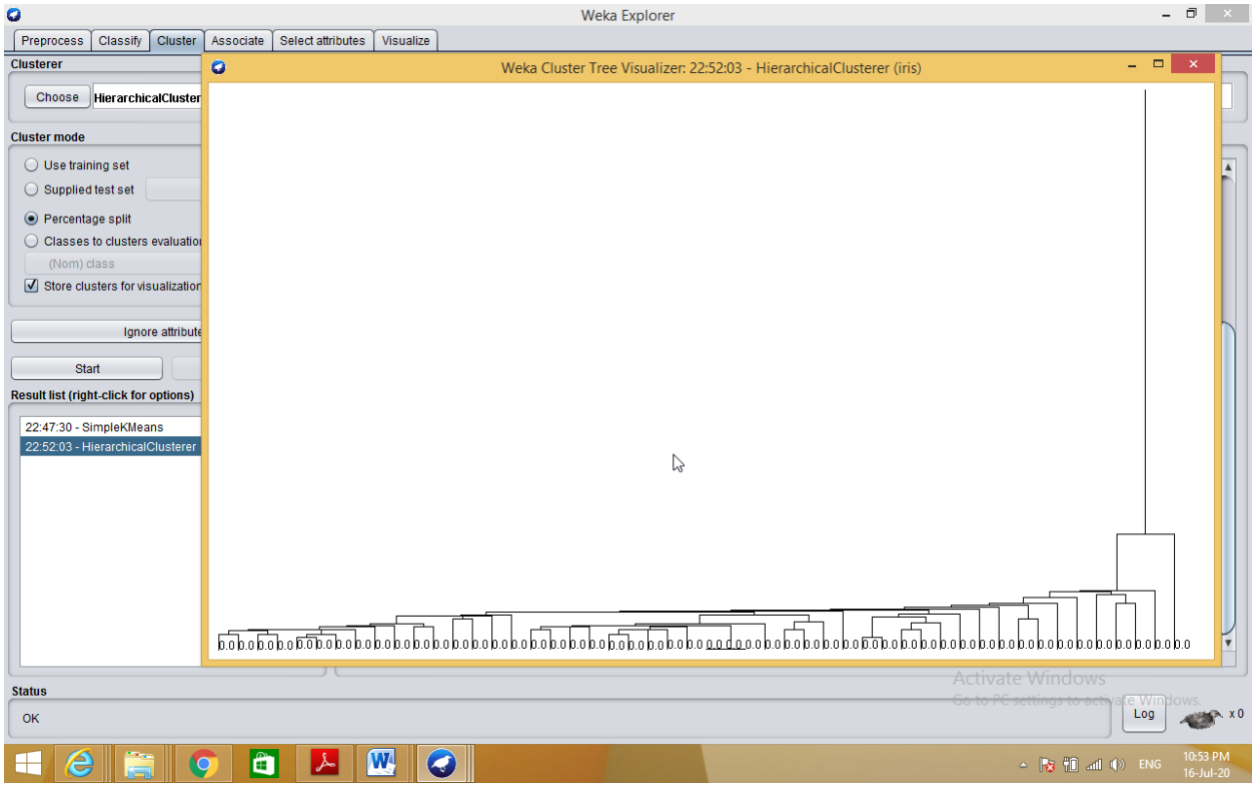
Step 2: Go to Cluster tab→choose→Hierarchical clustering

Dataset student .arff

```
@relation student
@attribute age {<30,30-40,>40}
@attribute income {low, medium, high}
@attribute student {yes, no}
@attribute credit-rating {fair, excellent}
@attribute buyspc {yes, no}
@data
%
<30, high, no, fair, no
<30, high, no, excellent, no
30-40, high, no, fair, yes
>40, medium, no, fair, yes
>40, low, yes, fair, yes
>40, low, yes, excellent, no
30-40, low, yes, excellent, yes
<30, medium, no, fair, no
<30, low, yes, fair, no
>40, medium, yes, fair, yes
<30, medium, yes, excellent, yes
30-40, medium, no, excellent, yes
30-40, high, yes, fair, yes
>40, medium, no, excellent, no
%
```


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
HierarchicalClusterer -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance -R first-last"
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Nom) class
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
22:47:30 - SimpleKMeans
22:52:03 - HierarchicalClusterer
Clusterer output
=== Clustering model (full training set) ===
Cluster 0
Cluster 1
Time taken to build model (full training data) : 0.09 seconds
=== Model and evaluation on test split ===
Cluster 0
Cluster 1
Time taken to build model (percentage split) : 0.02 seconds
Clustered Instances
0 36 (71%)
1 15 (29%)
Status
OK
Log
x 0
ENG
10:52 PM
16-Jul-20


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
HierarchicalClusterer -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance -R first-last"
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Nom) class
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
22:47:30 - SimpleKMeans
22:52:03 - HierarchicalClusterer
View in main window
View in separate window
Save result buffer
Delete result buffer(s)
Load model
Save model
Re-evaluate model on current test set
Re-apply this model's configuration
Visualize cluster assignments
Visualize tree
Clusterer output
=== Clustering model (full training set) ===
Cluster 0
Cluster 1
Time taken to build model (full training data) : 0.09 seconds
=== Model and evaluation on test split ===
Cluster 0
Status
OK
Log
x 0
ENG
10:52 PM
16-Jul-20


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
HierarchicalCluster
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters evaluatio
(Nom) class
Store clusters for visualizatio
Ignore attribute
Start
Result list (right-click for options)
22:47:30 - SimpleKMeans
22:52:03 - HierarchicalClusterer
Weka Cluster Tree Visualizer: 22:52:03 - HierarchicalClusterer (iris)
Status
OK
Log
x 0
Activate Windows
Go to PC settings to activate Windows.
ENG
10:53 PM
16-Jul-20